

UNIVERSITY OF SINDH



NAME

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ROLL NO

CSE/2K24/80

SUBJECT

HCI AND COMPUTER GRAPHIC

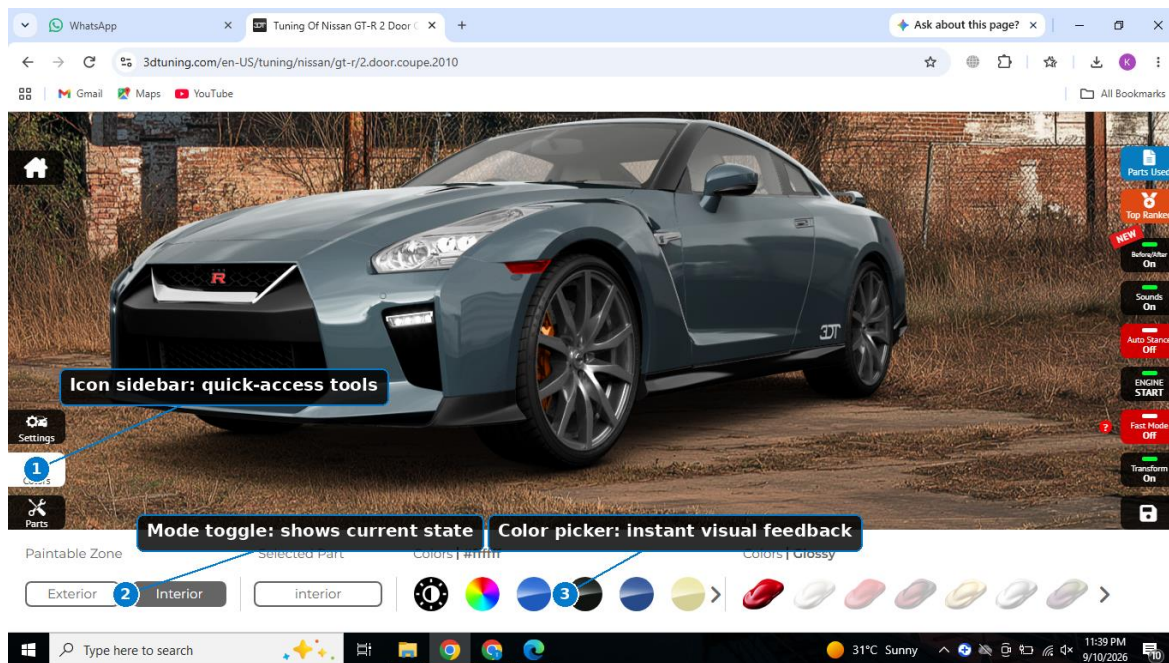
SUBMITTED TO

SIR RAJESH KUMAR

Mini Assignment: Deconstructing HCI & CG in Daily Software

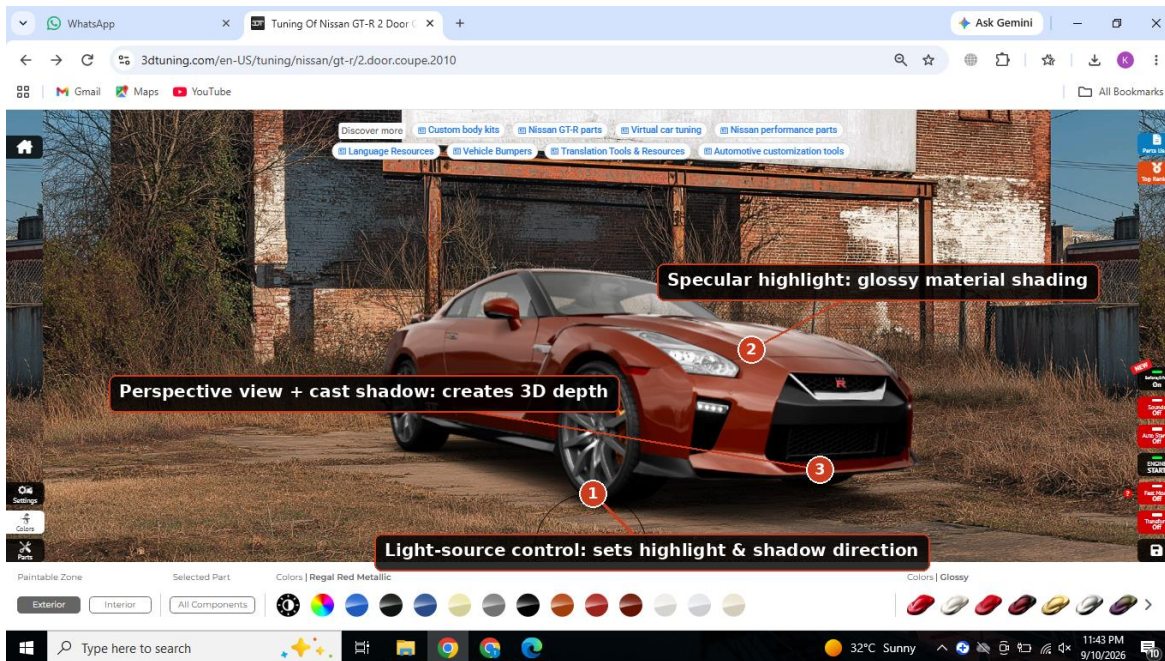
Tool Chosen: **3DTuning.com** — Online 3D Car Configurator (Nissan GT-R)

Slide / Image 1 — HCI Focus (Interaction Elements)



- 1. Icon sidebar (Settings / Colors / Parts):** A vertical row of icon buttons on the left lets the user quickly switch between customization modes without hunting through menus. The icon + label combo makes each tool's function clear at a glance.
- 2. Exterior / Interior toggle:** This pair of toggle buttons shows the user which paintable zone is currently active (the selected button is shaded dark). Clear state feedback like this prevents the user from being confused about what part of the car they are editing.
- 3. Color swatch picker:** Clicking a swatch instantly repaints the car in the 3D viewport. This gives immediate, direct visual feedback so the user can compare options without extra steps like clicking 'Apply'.

Slide / Image 2 — CG Focus (Rendering / Visual Elements)



- 1. Light-source control:** The tool exposes a draggable marker that sets the direction of the virtual sun. Moving it changes where highlights and shadows fall on the car, showing how a single light source drives the whole rendering.
- 2. Specular highlights on glossy paint:** The bright streak along the hood and windshield is a specular highlight produced by the glossy material shader reacting to the light source. It is what makes the paint read as smooth, reflective metal rather than a flat color fill.
- 3. Perspective camera view + cast shadow:** The car is rendered with a perspective camera (parallel lines converge toward the background) and casts a soft shadow onto the ground plane. Together these depth cues make the flat screen image read as a solid object sitting in 3D space.

Reflection Questions

- 1. How does the Computer Graphics quality help or hinder interaction?** The realistic lighting and glossy shading make it much easier to judge what a paint color or part will actually look like, which directly supports the user's decision-making. The trade-off is that a heavier 3D render can load and respond more slowly than a simple 2D image, which can make the interaction feel less snappy on a slower connection.
- 2. One HCI change to improve the interface:** I would add a short text label or tooltip next to the light-source dot, since right now it just looks like a stray marker on the ground and a new user would not immediately know it controls lighting direction.